

Nuclear Burning Stages of Stars: Hydrogen Burning and the $p+p$ Reaction

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Outline

Why Study Burning Stages?

Burning Stages: Temperature, Density, and Duration

Hydrogen Burning: Fusing 4 Protons into Helium-4

Reaction Rate, Lifetime, and the pp Chain

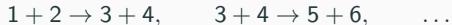
Calculating the Age of the Sun

Summary

Why Study Burning Stages?

Two Goals of Nuclear Astrophysics

A single isolated nuclear reaction is not enough to describe a star – reactions form a **network**:



Two driving questions:

- How is the **energy** produced inside stars, and which reactions are responsible?
- Which reactions are responsible for the **synthesis of elements** seen across the periodic table?

Governing principle: nuclides with the *lowest* Coulomb barrier present at a given time react first (highest tunneling probability). Once consumed, gravitational contraction raises the core temperature until the next-higher-barrier fuel ignites. This sequence defines the burning stages: hydrogen $\rightarrow$ helium $\rightarrow$ carbon $\rightarrow$ neon $\rightarrow$ oxygen $\rightarrow$ silicon, followed by neutron-capture *s*-, *r*-, (*p*-)processes for elements beyond iron.

Gamow Peak Shifts with Coulomb Barrier

At $T = 0.03$ GK (30 MK), comparing Gamow peaks for three reactions:

- $p + p$ (lowest V_c): peak at lowest energy
- $^{12}\text{C} + p$: peak shifts to higher energy
- $^{12}\text{C} + \alpha$: peak shifts even higher (strong $Z_1 Z_2$ dependence)

Takeaway: this directly confirms why nuclei with the lowest Coulomb barrier are consumed first within a star – their Gamow peak occurs at the lowest, most accessible energy. The existence of distinct burning stages is itself evidence that the Coulomb barrier governs stellar structure and evolution.

Burning Stages: Temperature, Density, and Duration

Central Temperature–Density Track ($M = 25 M_{\odot}$)

For a star of mass $25 M_{\odot}$ (chosen because this mass reproduces solar-system elemental abundances well), plotting central density vs. central temperature:

- Burning stages appear in order: $H \rightarrow He \rightarrow C \rightarrow Ne \rightarrow O \rightarrow Si$
- Central **temperature** varies by only about 2 orders of magnitude across all stages
- Central **density** varies by about 8 orders of magnitude

s - and r -process (neutron-induced) are not shown on this track – the diagram covers only charged-particle induced non-resonant and resonant burning stages.

Duration and Energy Yield of Burning Stages

Burning stage	Energy yield (MeV/g)	Typical duration
Hydrogen	$\sim 6 \times 10^{24}$	$10^6\text{--}10^{10}$ y
Helium	$\sim 6 \times 10^{23}$	shorter
Carbon	$\sim 3 \times 10^{23}$	shorter still
Oxygen	$\sim 5 \times 10^{17}$	much shorter
Silicon ($M = 25M_{\odot}$)	–	~ 1 day

- For H and He burning: almost all energy converts directly to detectable light
- For later stages: increasing loss to escaping neutrinos/antineutrinos, worsening with rising T
- $\approx 90\%$ of observed stars are caught burning hydrogen – later stages are simply too fast to catch in progress
- Elements with $A = 16\text{--}64$ form mainly in these advanced (post-H/He) burning stages; beyond the iron peak, synthesis is via neutron capture (s - and r -processes), not these charged-particle reactions

Hydrogen Burning: Fusing 4 Protons into Helium-4

The Net Reaction and Why It's Not One Step

Net hydrogen-burning reaction: $4p \rightarrow {}^4\text{He}$, with $Q = 26.731$ MeV (from mass excess values). **Why not a single 4-body collision?** Simultaneous interaction of 4 protons cannot account for the observed abundance of ${}^4\text{He}$ – such a process is statistically negligible. **Why not via ${}^2\text{He}$?** $p + p \rightarrow {}^2\text{He}$ produces a highly unstable intermediate, which cannot explain ${}^4\text{He}$ production either. $\Rightarrow$ Helium-4 is built via two distinct multi-step pathways:

- **pp chain** (proton–proton chain)
- **CNO cycle** (carbon–nitrogen–oxygen cycle)

Both operate over $T \approx 8\text{--}55$ MK (Sun's interior: ≈ 15 MK).

The Fundamental Step: $p + p \rightarrow d + e^+ + \nu$

This first step proceeds via the **weak interaction** only (not nuclear or EM) – because any nuclear/EM pathway through ${}^2\text{He}$ or ${}^5\text{Li}$ leads to highly unstable products.

$$p + p \rightarrow d + e^+ + \nu, \quad Q = 1.44 \text{ MeV (includes } e^+ \text{ annihilation energy)}$$

- Annihilation energy of e^+ : $2m_0c^2 = 1.02 \text{ MeV}$ (511 keV + 511 keV)
- Kinetic energy shared between e^+ and ν : $Q - 2m_0c^2 = 0.42 \text{ MeV}$

Alternate route – PEP reaction: $p + e^- + p \rightarrow d + \nu$ (same $Q = 1.44 \text{ MeV}$)

- Cross section is ~ 4 orders of magnitude *lower* than $p + p \Rightarrow$ negligible role in hydrogen burning
- But emits **mono-energetic** neutrinos at $E_\nu = 1.44 \text{ MeV}$ – potentially valuable for solar neutrino detection, since mono-energetic signals are easier to identify than the poly-energetic spectrum from the main $p + p$ channel

Energy Sharing: Density of States for e^+ and ν

Via Fermi's golden rule (details beyond this course), the energy distribution between the emitted positron and neutrino in $p + p \rightarrow d + e^+ + \nu$:

- Maximum available kinetic energy for either particle: $Q - 2m_0c^2 = 0.42 \text{ MeV}$
- **Average** neutrino energy: $\langle E_\nu \rangle \approx 0.263 \text{ MeV}$
- Average positron energy is correspondingly lower

Why this matters: stellar energy *loss* occurs because neutrinos/antineutrinos escape without interacting. To compute the net usable energy output of a star, the average neutrino energy carried away per reaction is essential – this number reappears in the age-of-the-Sun calculation ahead.

Why $p + p$ Cannot Be Measured in the Lab

Theoretical cross section at $E_p = 1$ MeV (lab frame): $\sigma \sim 10^{-47} \text{ cm}^2$. **Feasibility estimate:**

- Proton beam current: 1 mA
- Thick hydrogen target: $10^{23} \text{ atoms/cm}^2$
- Resulting interaction rate: **one interaction every 10^6 years**

Measuring a cross section needs thousands of events – utterly infeasible with any foreseeable technology (current, target density, background suppression). The $p + p$ reaction rate must instead be obtained *theoretically*.

Reaction Rate, Lifetime, and the pp Chain

Astrophysical S -Factor and Reaction Rate

Since both protons are charged, $p + p$ is a **non-resonant** reaction. Extrapolated S -factor at zero energy:

$$S(0) \approx 10^{-22} \text{ keV} \cdot \text{b}$$

At $T_6 = 15$ (solar interior):

$$\langle \sigma v \rangle_{pp} \approx 10^{-43} \text{ cm}^3 \text{s}^{-1}$$

Mean lifetime of a proton against $p + p$ fusion:

$$\tau = \frac{1}{N_H \langle \sigma v \rangle_{pp}}$$

With $X_H = X_{He} = 0.5$ and $\rho = 100 \text{ g/cm}^3$: $\tau \approx 10^{10}$ years – comparable to the age of the oldest known stars. **This is why stars burn hydrogen so slowly:** the $p + p$ step is the bottleneck of the entire chain.

Completing the pp I Chain

Once deuterium is formed, the chain proceeds rapidly:



This is the **pp I chain**: overall 4 protons in, ${}^4\text{He}$ out, with 2 protons released back into the plasma.

- These two steps proceed via **electromagnetic and nuclear** interactions – much faster than the weak-interaction $p + p$ step
- Because they are fast, they do not limit the chain
- $\Rightarrow$ the *overall* pp-chain reaction rate is governed entirely by the slow $p + p$ step

Calculating the Age of the Sun

Setting Up the Calculation

Net process: $4p \rightarrow {}^4\text{He} + 2e^+ + 2\nu$

- Energy lost to 2 escaping neutrinos: $2 \times 0.263 = 0.526 \text{ MeV}$
- Effective energy retained by the star: $26.73 - 0.53 = 26.2 \text{ MeV}$ per fusion of 4 protons

Solar surface luminosity: $L_{\odot} \approx 2.4 \times 10^{39} \text{ MeV/s}$ **Rate of fusion processes:**

$$N = \frac{L_{\odot}}{26.2 \text{ MeV}} \approx 10^{38} \text{ processes/s}$$

Each process consumes 4 protons $\Rightarrow$ mass of hydrogen converted to helium per second:

$\approx 616 \text{ million tonnes of H} \rightarrow \text{He, every second}$

From Fuel-Consumption Rate to Stellar Age

Using $M_{\odot} = 2 \times 10^{30}$ kg and the consumption rate above, assuming a *uniform* processing rate of all the Sun's hydrogen:

$$\text{upper limit on lifetime} \approx 10^{11} \text{ years}$$

Correction: burning actually occurs only in the core, and involves just $\approx 10\%$ of the Sun's total hydrogen:

$$\text{lifetime} \approx 10^{11} \times 0.1 = 10^{10} \text{ years}$$

Compare to the Sun's observed present age: 4.5×10^9 years. **Conclusion:** the Sun is roughly midway through its hydrogen-burning lifetime – consistent with it being one of the many **middle-aged** stars in the galaxy.

Summary

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- Burning stages follow the Coulomb-barrier ordering: $H \rightarrow He \rightarrow C \rightarrow Ne \rightarrow O \rightarrow Si$, then s -/ r -process beyond iron; governed by a $25M_{\odot}$ reference model matching solar abundances
- Temperature varies by $\sim 10^2$, density by $\sim 10^8$ across the burning sequence; later stages (e.g. Si: ~ 1 day) are vastly shorter than hydrogen burning (10^6 – 10^{10} y) – hence $\approx 90\%$ of observed stars are caught burning hydrogen
- $4p \rightarrow {}^4He$ proceeds via the **pp chain** (and separately the CNO cycle); the rate-limiting weak-interaction step $p + p \rightarrow d + e^+ + \nu$ has $\sigma \sim 10^{-47} \text{ cm}^2$ at 1 MeV – unmeasurable in any lab, requiring theoretical extrapolation of $S(0)$
- Average neutrino energy (≈ 0.263 MeV) sets the energy lost from the star per reaction; this allows the famous "age of the Sun" estimate ($\approx 10^{10}$ y for full H consumption; $\approx 10^{10}$ y corrected for 10% core burning), consistent with the Sun's true age of 4.5×10^9 y